

1. PRELIMINARIES

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

Definition 1.2 (Central factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right), & \text{if } k > 0. \end{cases}$$

Proposition 1.3 (Central Newton's formula). *For arbitrary integers x , and t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes central finite difference of $f(t)$, defined by

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right).$$

Proof. See [?, p. 462]. □

Lemma 1.4 (Central falling factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0. \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right)$ are falling factorials.

Lemma 1.5 (Binomial form of central factorials). *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left(n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k(k-1)!} \left(n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$, and Lemma (1.4). $\square$

Lemma 1.6 (Halved central factorials). *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \binom{n + \frac{k}{2}}{k} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n + \frac{k}{2}}{k} = \frac{n + \frac{k}{2}}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Thus,

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \binom{n + \frac{k}{2}}{k} - \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[2 \binom{n + \frac{k}{2}}{k} - \binom{n + \frac{k}{2} - 1}{k-1} \right] = \frac{1}{2} \left[\binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} + \binom{n + \frac{k}{2} - 1}{k-1} - \binom{n + \frac{k}{2} - 1}{k-1} \right].$$

Thus, the statement follows. $\square$

Proposition 1.7 (Central Newton's formula for powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Proposition 1.8 (Halved central factorial powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t + \frac{k}{2}}{k} + \binom{n-t + \frac{k}{2} - 1}{k} \right] \delta^k t^m$$

Proposition 1.9 (Generalized hockey-stick identity). *For integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. $\square$

Lemma 1.10 (Centered hockey-stick identity). *For integers $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $j = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k}.$$

By Generalized hockey-stick identity (1.9),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows. $\square$

Proposition 1.11 (Ordinary sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \delta^k t^m \sum_{j=1}^n \left[\binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k} \right]$$

Proposition 1.12 (Ordinary sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] \delta^k t^m$$

Proposition 1.13 (Central factorial numbers sums). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n + \frac{k}{2} + 1}{k+1} - \binom{1 + \frac{k}{2}}{k+1} + \binom{n + \frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1} \right] k! T(m, k),$$

where $T(n, k)$ denote Central factorial numbers of the second kind (in Riordan's notation), defined by polynomial identity,

$$x^m = \sum_{k=1}^m T(m, k) x^{[k]}.$$

See [?, p. 213], and [?].

Definition 1.14 (Generalized Central factorial numbers of the second kind). *For integers $0 \leq k \leq m$, and an arbitrary integer t ,*

$$T_t(m, k) = \frac{\delta^k t^m}{k!}.$$

Proposition 1.15 (Generalized Central factorial sums). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n - t + \frac{k}{2} + 1}{k+1} - \binom{1 - t + \frac{k}{2}}{k+1} + \binom{n - t + \frac{k}{2}}{k+1} - \binom{-t + \frac{k}{2}}{k+1} \right] k! T_t(m, k)$$

Proposition 1.16 (Double sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^m \left[+ \binom{n - t + \frac{k}{2} + 2}{k+2} - \binom{2 - t + \frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{1 - t + \frac{k}{2}}{k+1} \Sigma^1 n^0 \right. \\ \left. + \binom{n - t + \frac{k}{2} + 1}{k+2} - \binom{1 - t + \frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{0 - t + \frac{k}{2}}{k+1} \Sigma^1 n^0 \right] \delta^k t^m \end{aligned}$$

Proposition 1.17 (Triple sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^3 n^m = \frac{1}{2} \sum_{k=0}^m \left[\right. \\ + \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \\ \left. + \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \right] \delta^k t^m \end{aligned}$$

Proposition 1.18 (Multifold sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k t^m \end{aligned}$$

2. MATHEMATICA

Mathematica Function	Validates / Prints
<code>MultifoldSumOfPowersRecurrence[r, n, m]</code>	Computes $\Sigma^r n^m$, (1.1)
<code>CentralFactorial[n, k]</code>	Computes $n^{[k]}$, (1.2)
<code>ValidateCentralFactorialsInTermsOfFalling[10]</code>	Validates Lem. (1.4)
<code>ValidateHalvedCentralBinomials[20]</code>	Validates Lem. (1.6)
<code>ValidateNewtonsFormulaForPowers[10]</code>	Validates Prop. (1.7)
<code>ValidateHalvedCentralFactorialPowers[5]</code>	Validates Prop. (1.8)
<code>ValidateCenteredHockeyStickIdentity[5]</code>	Validates Lem. (1.10)
<code>ValidateOrdinarySumsOfPowersDecomposition[10]</code>	Validates Lem. (1.12)
<code>ValidateDoubleSumsOfPowersDecomposition[5]</code>	Validates Lem. (1.16)
<code>ValidateTripleSumsOfPowersDecomposition[5]</code>	Validates Lem. (1.17)
<code>ValidateMultifoldSumsOfPowers[3]</code>	Validates Lem. (1.18)